Nishal Stanislaus Silva, *PhD*

Machine Learning Infrastructure Engineer | Real-Time Inference, End-to-End Pipeline Ownership, Ranking & Embeddings

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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14th October 2026

Re: Senior Machine Learning Infrastructure Engineer, Embedding Platform – Reddit

Dear Hiring Team,

Your posting asks for someone who can own ML components end to end, from problem definition through production deployment, and build reliable serving for low-latency, high-throughput systems. That is the job I have done for the last decade, most recently on the EU-funded MUSMET project at the University of Trento, where I owned a full ML pipeline, acquisition through monitoring, for streaming sensor and audio data.

Two results from that work speak to what you are building. The inference services I shipped ran at 14 ms latency and F1 0.76 on edge hardware, a tighter constraint than most cloud ranking systems operate under. And a benchmark suite I designed to test architecture and feature trade-offs led to a model 74× faster than the existing baseline while improving accuracy, the same baselines-plus-ablations discipline your team uses for offline and online evaluation. Before the PhD I led ML and computer-vision development at MAS Holdings, building real-time ETL pipelines processing millions of IoT events per day across distributed sites and writing the C++ and Python inference engines that ran on embedded hardware in production, which taught me the difference between a model that works in a notebook and a system that survives real load, partial failures, and stakeholders who need to trust it.

I do not have direct experience training embedding models at internet scale, and I would rather say that plainly than let the resume imply otherwise. What I bring instead is the pipeline-ownership and latency-optimization instinct that scale problems are built on, plus a research background that treats evaluation as a first-class engineering problem. Reddit's embedding platform sits under recommendation, search and safety at once, and I would expect to contribute on serving reliability and evaluation tooling early.

I am a Canadian permanent resident based in Toronto, eligible to work without sponsorship, open to remote work, and available immediately. I would welcome the chance to discuss how this experience maps onto the Embedding Platform team's work. My portfolio and publications are at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD